SARTHAK SRIVASTAVA

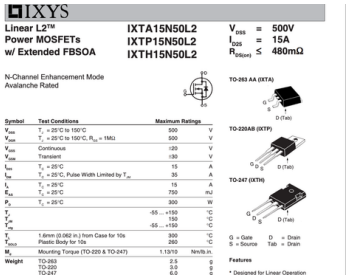
MECHATRONICS ENGINEERING STUDENT

ELECTRONIC LOAD BOX | SOLACE POWER



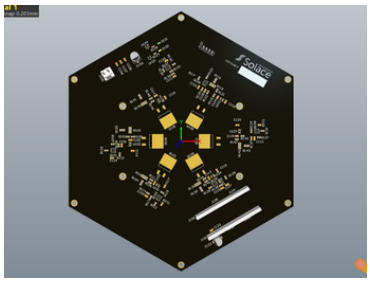
What

- Developed a compact, portable electronic DC load for internal testing as an alternative to a commercial load box.
- Designed the system around power MOSFETs operating in the linear region for controlled power dissipation.



How

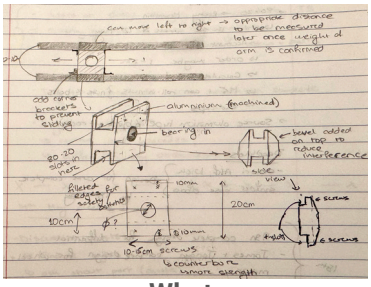
- Used LTspice to simulate MOSFET power dissipation and performance.
- Referenced manufacturer datasheets to verify ratings and safe operating limits.
- Evaluated MOSFET behaviour in the linear region under expected load conditions.



Results

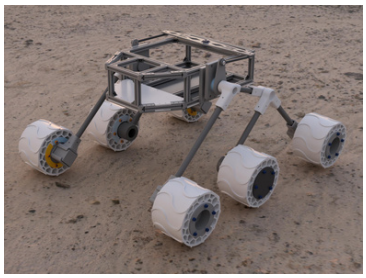
- Designed and manufactured a functional PCB in Altium for internal electronic testing.
- Designed for continuous operation up to 150 W, with the capability to handle up to 400 W under appropriate operating conditions.

CHASSIS-SUSPENSION DESIGN | SIDUS ROBOTICS



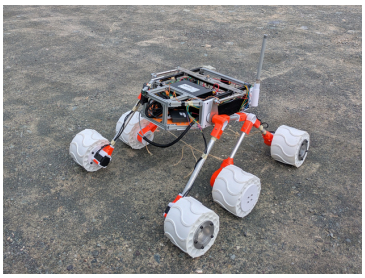
What

- Lead design and development of the chassis & suspension for Mars Rover
- Designed, mentored team members and managed mechanical design for chassis - suspension
- Goal was to create a lightweight and sturdy rover while minimizing costs



How

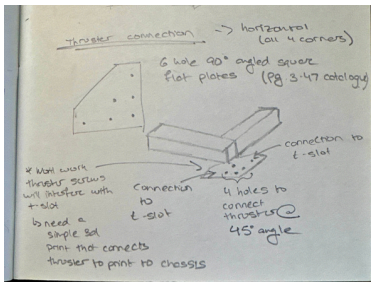
- Researched existing Mars Rover designs
- Collaborated with electrical and software team to create designs based on equipment chosen
- Researched materials for a strong and lightweight design



Results

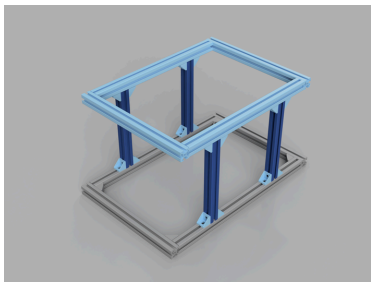
- Completed 1st prototype of the Mars Rover
- Successful drive test conducted

CHASSIS AND THRUSTER MOUNT DESIGN | EASTERN EDGE ROBOTICS



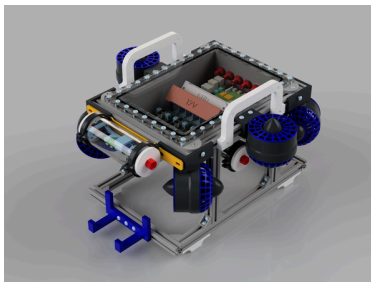
What

- Designed the chassis and thruster mounts for Eastern Edge Robotics' ROV.
- Chassis design prioritized compactness, low weight, and structural strength.
- Thruster mounts were required to be compact and capable of withstanding multi-axial loading.



How

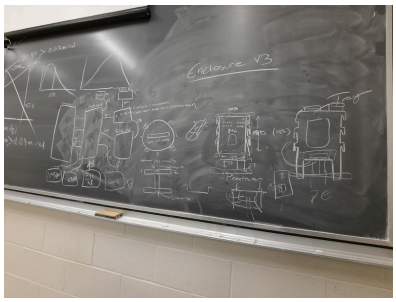
- Adapted and improved upon a chassis design from the previous year's ROV to meet updated system requirements.
- Engineered thruster mounts to securely hold the motors, using 45-degree 3D printing orientation to enhance strength under multi-directional forces.



Results

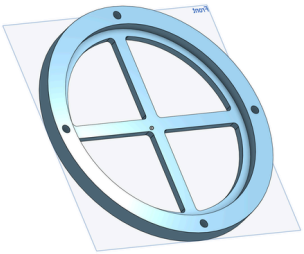
- Delivered a modular chassis system that supports custom tooling and mission-specific configurations.
- Final assembly allowed for quick deployment and field modifications, significantly improving ease of use and maintenance.

ELECTRONICS ENCLOSURE | EASTERN EDGE ROBOTICS



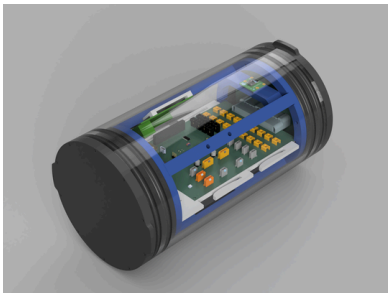
What

- Designed an enclosure to house all PCBs and wiring, focusing on streamlined assembly and efficient space utilization
- Prioritized ease of assembly and optimal internal layout to ensure effective organization and accessibility of components



How

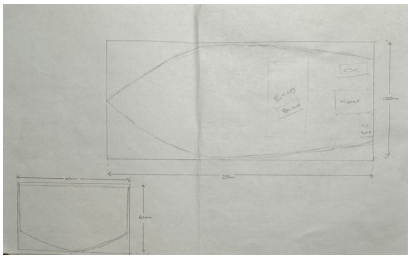
- Utilized OnShape to design all components with precision and functionality.
- Collaborated with the team to test fit, refine, and enhance the design for optimal performance.
- Conducted thorough research to ensure efficient space utilization and weight management in the final design.



Results

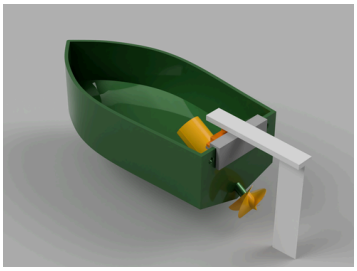
- Developed the third iteration of the enclosure, refining the design for improved functionality
- Prototyped the enclosure using 3D printing and conducted assembly tests to evaluate space management and identify areas for further optimization

REMOTE-CONTROL BOAT | THE FLUVARIUM



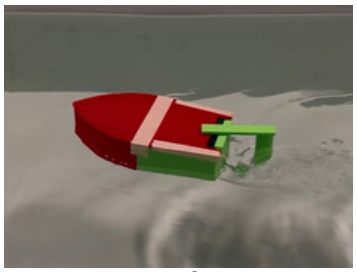
What

- Conceptualized and designed a remote-controlled boat specifically for competing in the annual rubber duck race hosted by the Fluvarium.
- Developed and manufactured a functional prototype to validate the feasibility and performance of the design.



How

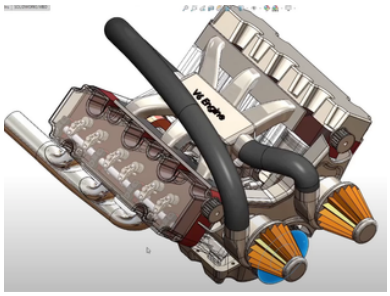
- Consulted with a professor to gain insights and guidance on boat design principles.
- Created initial sketches to outline constraints and develop basic design concepts on paper.
- Translated preliminary sketches into detailed CAD models using OnShape to finalize the design.
- Researched and selected electronics for the remote-control system, designing components to integrate with them.



Results

- 3D printed and assembled the final prototype based on iterative improvements.
- Tested the prototype for water leakage using insights from earlier iterations.
- Conducted successful performance testing in a controlled indoor water environment, ensuring efficient operation.

V6 ENGINE DESIGN | PERSONAL PROJECT



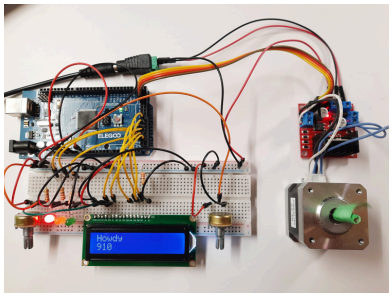
What

- Designed a detailed V6 engine model to gain proficiency in SolidWorks and deepen understanding of engine mechanics, parts, and assembly
- Created a functional 3D-printed model powered and controlled by an Arduino, demonstrating technical skills in both design and electronics integration



How

- Followed a detailed online tutorial with CAD drawings to guide the design process
- Created individual parts and developed both sub-assemblies and full assemblies
- Researched components and techniques for controlling the model, incorporating a stepper motor and potentiometer for precise operation



Results

- 3D printed and assembled individual engine components into a fully functional model
- Successfully completed Arduino circuitry for powering and controlling the engine